PROGRAM:

```
#include<stdio.h>
#include<limits.h>
void insert(int ary[], int hFn, int size)
{ int element, pos, n = 0;
printf("Enter key element to insert\n");
scanf("%d", &element);
pos = element % hFn;
while (ary[pos] != INT_MIN) {
// INT_MIN and INT_MAX indicate that cell is empty or
deleted if (ary[pos] == INT_MAX)
break;
```



```
pos = (pos + 1) % hFn;
n++;
```



```
if (n == size) {
printf("Hash table is full. No place to insert this
element.\n"); return;
```

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```
if (n == size) {
printf("Hash table is full. No place to insert this
element.\n"); return;
}
}
ary[pos] = element; // Inserting element
}
void delete(int ary[], int hFn, int size) {
int element, pos, n = 0;
printf("Enter element to delete\n");
scanf("%d", &element);
pos = element %
hFn; while (n++ !=
size) {
if (ary[pos] == INT_MIN) {
printf("Element not found in hash
table\n"); return;
} else if (ary[pos] == element) {
ary[pos] = INT_MAX; // Mark as deleted
printf("Element deleted\n");
return;
} else {
pos = (pos + 1) % hFn;
}
}
printf("Element not found in hash table\n");
}
void search(int ary[], int hFn, int size) {
int element, pos, n = 0;
printf("Enter element you want to search\n");
scanf("%d", &element);
pos = element %
hFn; while (n++ !=
size) {
if (ary[pos] == element) {
printf("Element found at index %d\n", pos);
return;
} else if (ary[pos] == INT_MAX || ary[pos] != INT_MIN) {
pos = (pos + 1) % hFn;
}
}
printf("Element not found in hash table\n");
}
```



```
void display(int ary[], int size) {
printf("Index\tValue\n");
for (int i = 0; i < size; i++) { printf("%d\t",
 i);
if (ary[i] == INT_MIN) {
```

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```
} else if (ary[i] == INT_MAX) {
printf("Deleted\n");
} else { printf("%d\n",
 ary[i]);
}
}
}
int main() {
int size, hFn, choice;
printf("Enter size of hash table\n");
scanf("%d", &size);
int ary[size];
printf("Enter hash function (if mod 10, enter 10)\n");
scanf("%d", &hFn);
for (int i = 0; i < size; i++)
ary[i] = INT_MIN; // Assigning INT_MIN indicates that cell is
empty do {
printf("\nEnter your choice\n");
printf("1 -> Insert\n2 -> Delete\n3 -> Display\n4 -> Search\n0 -> Exit\n");
scanf("%d", &choice);
switch (choice) {
case 1:
insert(ary, hFn, size);
break;
case 2:
delete(ary, hFn,
size); break;
case 3:
display(ary,
size); break;
case 4:
search(ary, hFn, size);
break;
case 0:
printf("Exiting...\n");
break;
default:
printf("Enter correct choice\n");
break;
}
} while (choice != 0);
return 0;
}
```

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